

Dependencies & Requirements

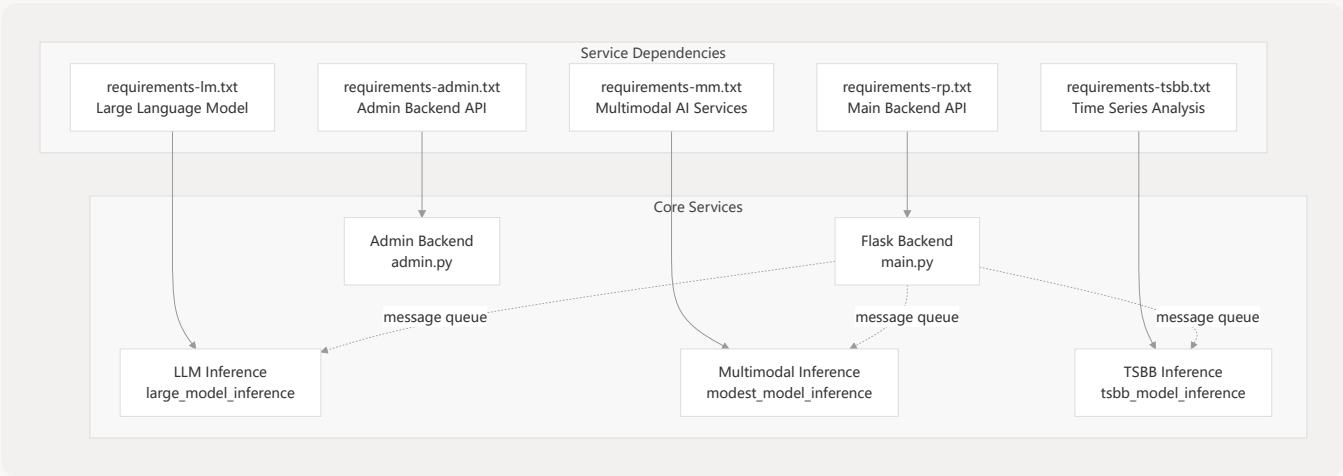
> Relevant source files

This document covers the Python package dependencies and system requirements for all components of the MKTY Smart Healthcare System. Each service has its own requirements file defining the specific packages needed for operation.

For information about system deployment and configuration, see [Deployment & Configuration](#). For details about the AI model services architecture, see [AI & Machine Learning Services](#).

Dependency Structure Overview

The MKTY system is organized into multiple services, each with dedicated dependency requirements files:



- Sources:** backend/requirements-rp.txt 1-17 backend/requirements-admin.txt 1-4
- backend/requirements-lm.txt 1-4 backend/requirements-mm.txt 1-7
- backend/requirements-tsbb.txt 1-4

Core Backend Dependencies

The main Flask backend API uses **requirements-rp.txt** and includes dependencies for web services, authentication, database connectivity, and document processing:

Package	Version	Purpose
Flask	3.0.3	Web framework for REST API

Package	Version	Purpose
Flask-Cors	5.0.0	Cross-origin resource sharing
Flask-JWT-Extended	4.7.1	JWT token authentication
gevent	24.11.1	Asynchronous web server
pika	1.3.2	RabbitMQ message queue client
mysql-connector-python	9.2.0	MySQL database connectivity
jieba	0.42.1	Chinese text segmentation
Pillow	10.4.0	Image processing
argon2-cffi	21.3.0	Password hashing
markdown	3.4.1	Markdown document processing
weasyprint	65.1	PDF generation
python-docx	1.1.2	Word document processing
python-pptx	1.0.2	PowerPoint processing
PyPDF2	3.0.1	PDF document processing
rank-bm25	0.2.2	Text search and ranking
numpy	1.26.4	Numerical computing
rich	13.7.1	Terminal output formatting

Sources: backend/requirements-rp.txt 1-17

AI Service Dependencies

Large Language Model Service

The requirements-lm.txt defines dependencies for the MKTY-3B-Chat language model service:

Package	Version	Purpose
torch	2.6.0	PyTorch deep learning framework
transformers	4.48.3	Hugging Face model loading
rich	13.7.1	Terminal output formatting
pika	1.3.2	RabbitMQ message queue client

Sources: backend/requirements-lm.txt 1-4

Multimodal AI Services

The requirements-mm.txt supports BioMedCLIP and MarianMT models for medical image analysis and translation:

Package	Version	Purpose
torch	2.6.0	PyTorch deep learning framework
Pillow	10.4.0	Image processing for medical images
open-clip-torch	2.23.0	BioMedCLIP vision-language model
transformers	4.48.3	MarianMT translation model
rich	13.7.1	Terminal output formatting
pika	1.3.2	RabbitMQ message queue client
numpy	1.26.4	Numerical array operations

Sources: backend/requirements-mm.txt 1-7

Time Series & Text Analysis

The requirements-tsbb.txt supports BigBird and GRU models for time series prediction and text analysis:

Package	Version	Purpose
torch	2.6.0	PyTorch deep learning framework
transformers	4.48.3	BigBird transformer model
rich	13.7.1	Terminal output formatting
pika	1.3.2	RabbitMQ message queue client

Sources: backend/requirements-tsbb.txt 1-4

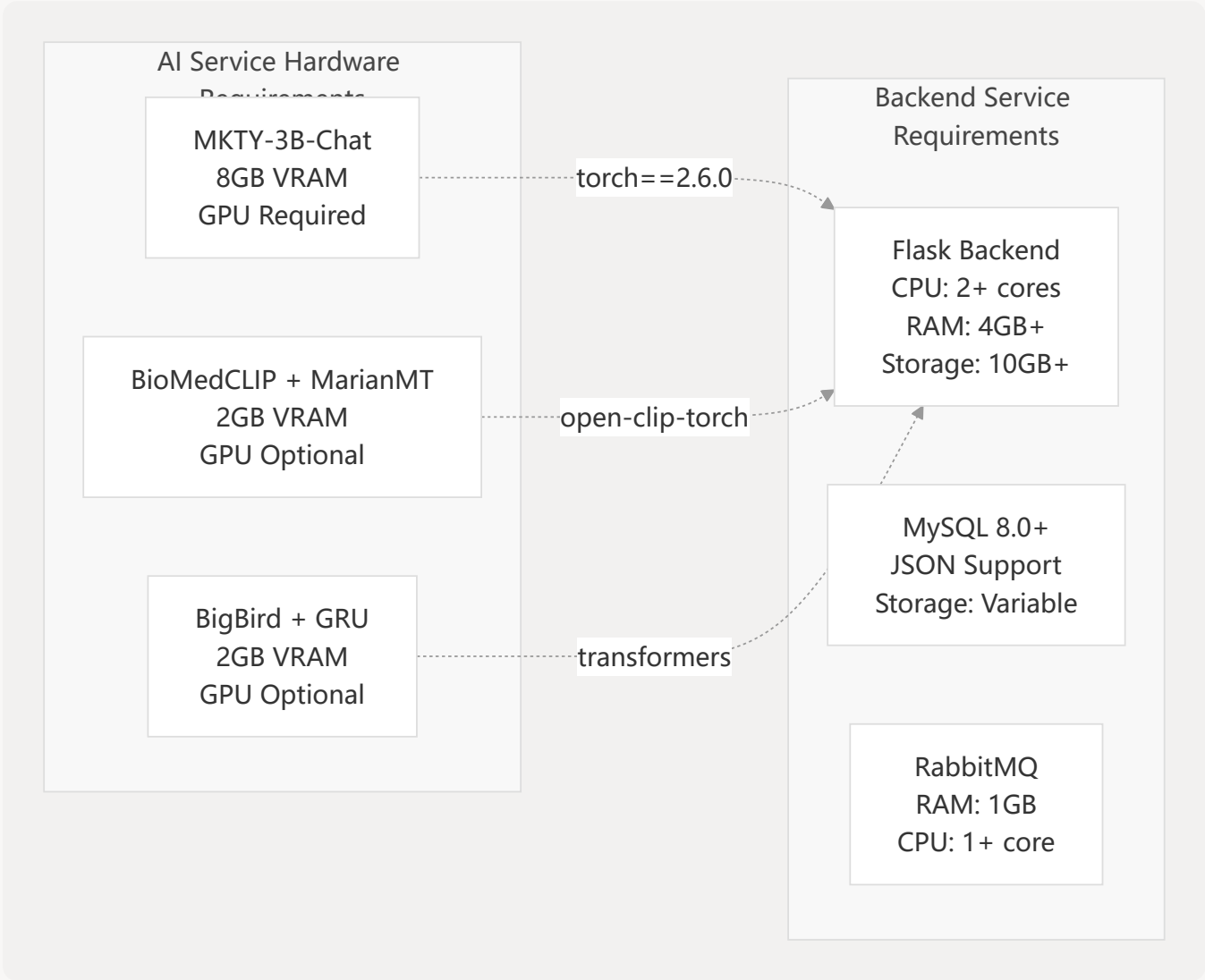
Administrative System Dependencies

The requirements-admin.txt defines minimal dependencies for the administrative backend API:

Package	Version	Purpose
Flask	3.0.3	Web framework
Flask-SQLAlchemy	3.1.1	ORM for database management
Flask-Cors	5.0.0	Cross-origin resource sharing
PyJWT	2.8.0	JWT token handling

Sources: backend/requirements-admin.txt 1-4

Hardware and System Requirements



Minimum System Requirements

- Operating System:** Linux (Ubuntu 20.04+ recommended), macOS, or Windows 10+
- Python Version:** 3.8+ (3.10+ recommended)
- CPU:** 4+ cores for full system deployment
- RAM:** 16GB+ for complete system with all AI services

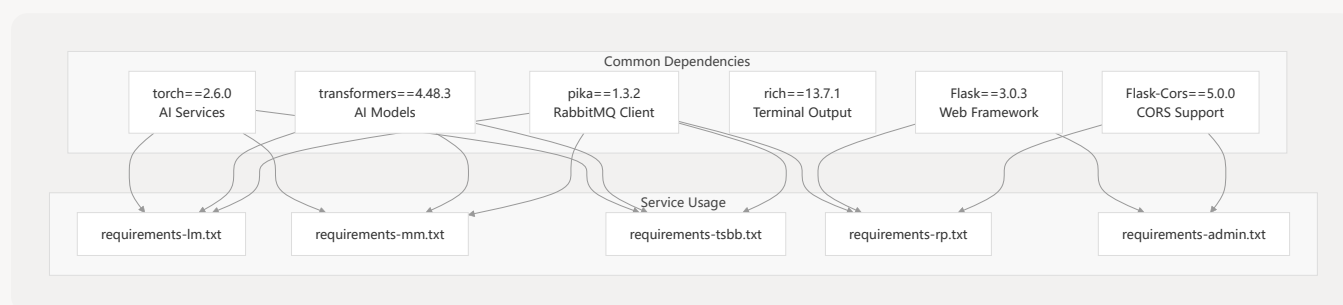
- **Storage:** 50GB+ for models, database, and application files
- **GPU:** NVIDIA GPU with 8GB+ VRAM for optimal AI performance

Database Requirements

- **MySQL:** Version 8.0 or higher with JSON datatype support
- **Storage:** Minimum 10GB, recommended 100GB+ for production
- **Connections:** Support for concurrent connections from multiple services

Sources: backend/requirements-rp.txt 6 backend/requirements-admin.txt 2

Shared Dependencies Analysis



The dependency analysis reveals several shared packages across services:

- **PyTorch (torch==2.6.0):** Used by all AI inference services for model loading and execution
- **Transformers (transformers==4.48.3):** Required for MKTY-3B-Chat, BioMedCLIP, BigBird, and MarianMT models
- **Pika (pika==1.3.2):** RabbitMQ client library used by main backend and all AI services for asynchronous communication
- **Flask (Flask==3.0.3):** Web framework for both main backend and admin APIs
- **Rich (rich==13.7.1):** Terminal output formatting used across AI services for logging

Sources: backend/requirements-lm.txt 1-4 backend/requirements-mm.txt 1-7

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